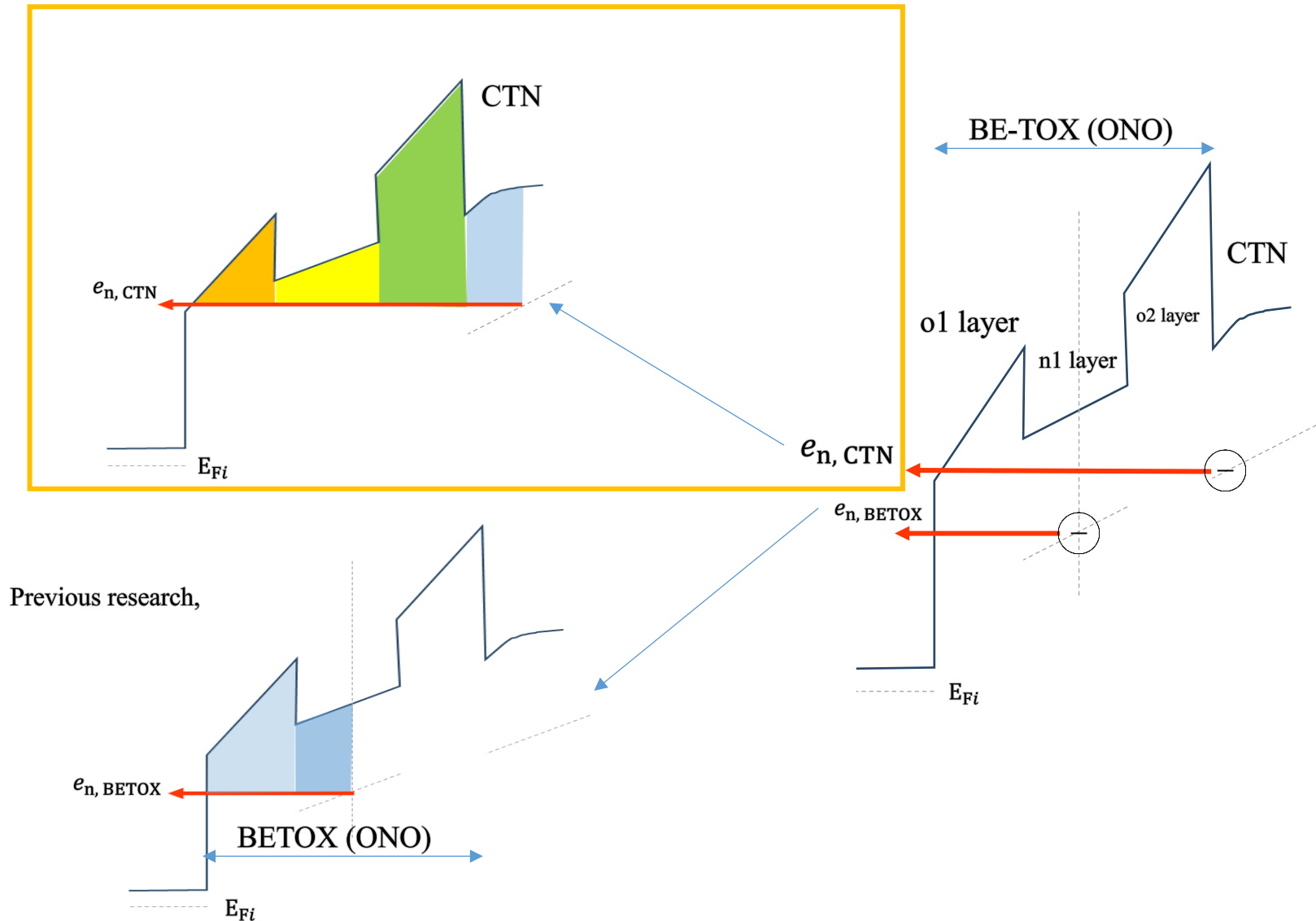


Modeling of CTN Detrapping

25년 9-11월 연구 정리

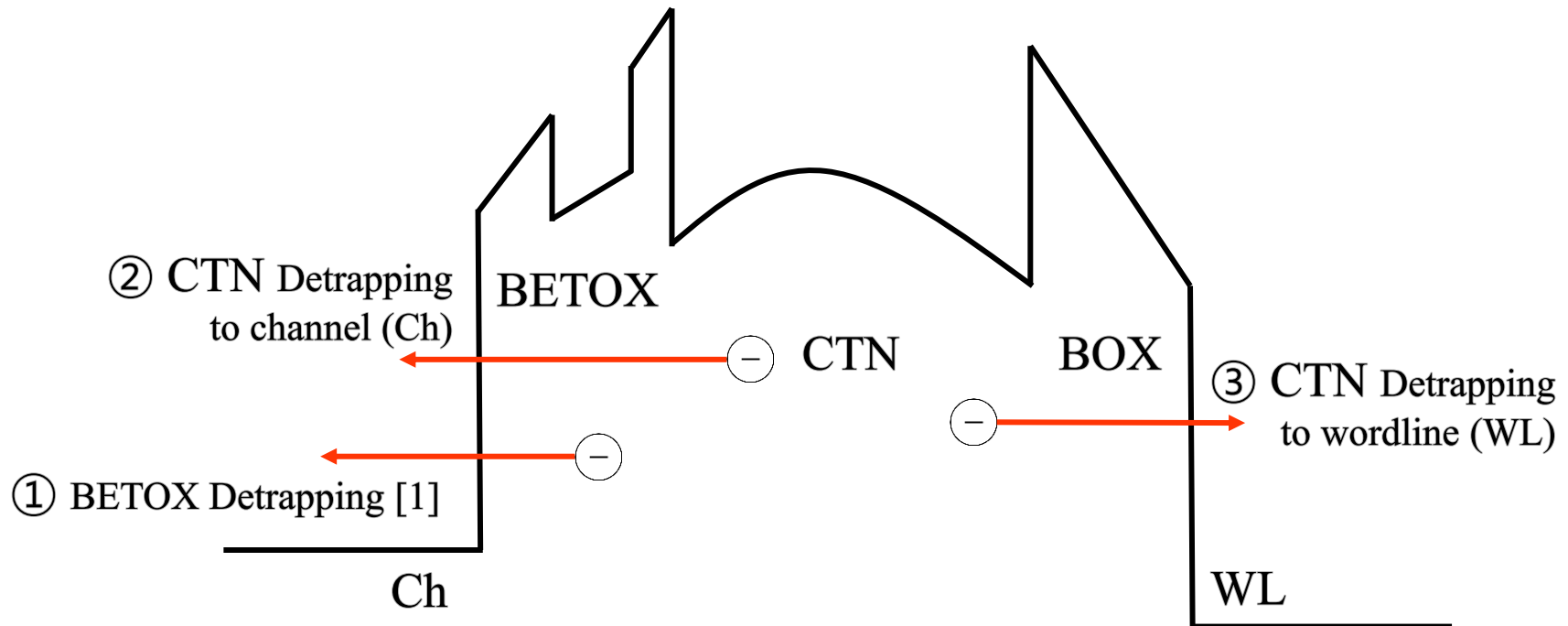
진명

Detrapping from CTN Idea



CTN Trapped Charge Detrapping

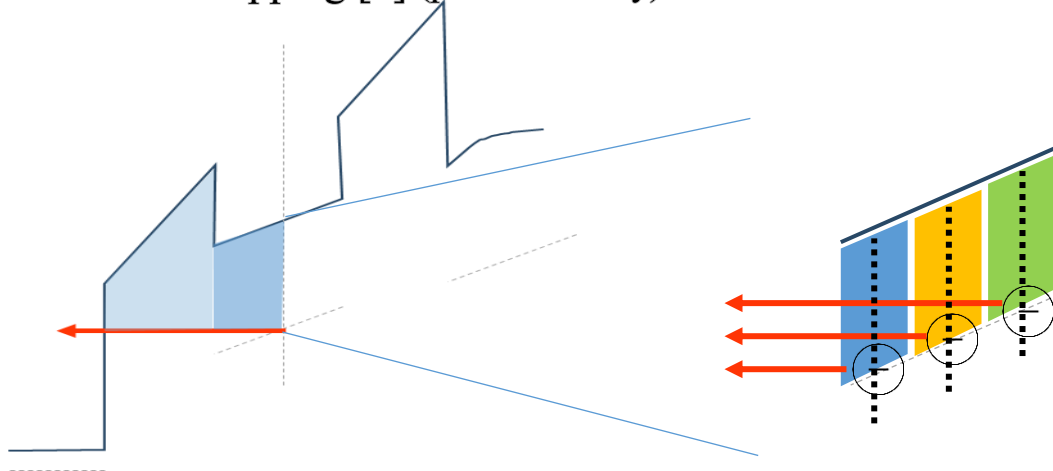
Detrapping mechanisms in retention of 3-D NAND flash vertical cross section



[1] Jin, TED, 2025

CTN Electron Emission Simulation Method

① BETOX Detrapping [1] (previous study)

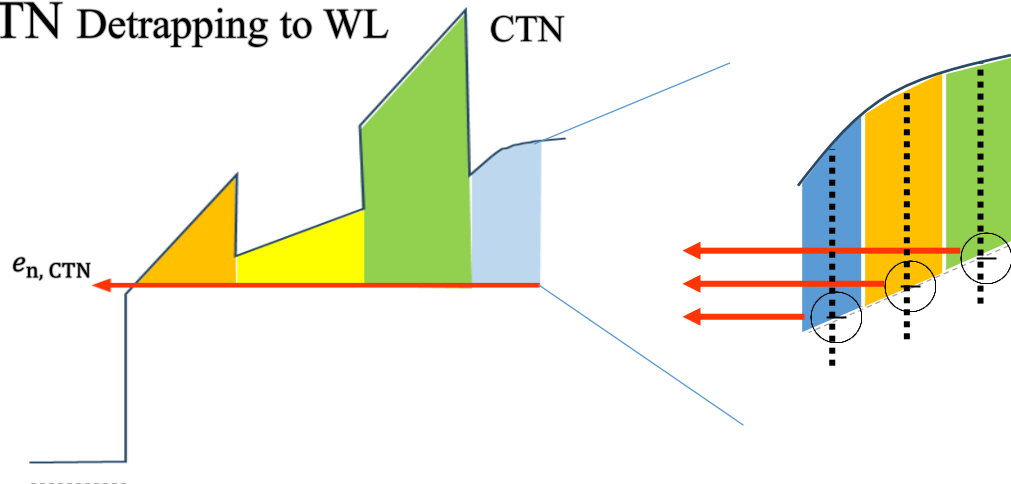


From Last Research...

Vertical Partition Method
=> Uniform field assumption

② CTN Detrapping to Ch

③ CTN Detrapping to WL

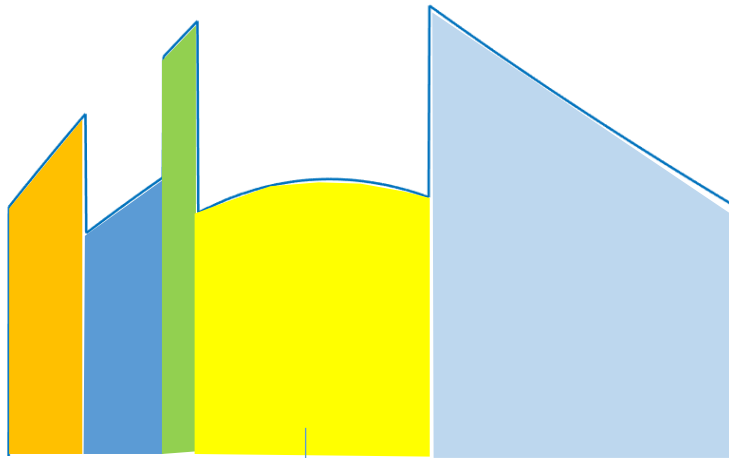


Same Implementation

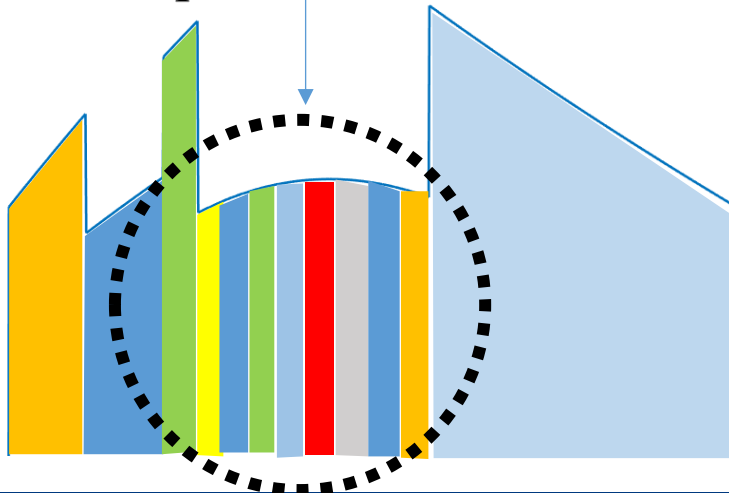
Vertical Partition Method
=> Non-uniform field!

CTN Model Poisson in Partitions

Original method: only a block of uniformly distributed trap model



New method: partitioned CTN model



Potential Solution:

Radial Field Intensity Solution:

$$\begin{aligned} V_{o1}(r) &= \frac{K}{\epsilon_{ox}} \ln \frac{r}{r_{Si}} + V_{ch} \\ V_{n1}(r) &= V_{o1}(r_{o1}) + \frac{K_1}{\epsilon_{n1}} \ln \frac{r}{r_{o1}} + \frac{qn_L n_1}{4\epsilon_{n1}} (r^2 - r_{o1}^2) \\ V_{o2}(r) &= V_{n1}(r_{n1}) + \frac{K_2}{\epsilon_{ox}} \ln \frac{r}{r_{n1}} \\ V_{CTN}(r) &= V_{o2}(r_{o2}) + \frac{K_3}{\epsilon_{CTN}} \ln \frac{r}{r_{o2}} + \frac{qn_L CTN}{4\epsilon_{CTN}} (r^2 - r_{o2}^2) \\ V_{BOX}(r) &= V_{CTN}(r_{CTN}) + \frac{K_4}{\epsilon_{ox}} \ln \frac{r}{r_{CTN}} = V_G - \frac{K_4}{\epsilon_{ox}} \ln \frac{r_{BOX}}{r} \end{aligned} \quad \begin{aligned} F_{o1}(r) &= -\frac{K}{\epsilon_{ox}} \frac{1}{r} \\ F_{n1}(r) &= -\frac{K_1}{\epsilon_{n1}} \frac{1}{r} - \frac{qn_L n_1}{2\epsilon_{n1}} r \\ F_{o2}(r) &= -\frac{K_2}{\epsilon_{ox}} \frac{1}{r} \\ F_{CTN}(r) &= -\frac{K_3}{\epsilon_{CTN}} - \frac{qn_L CTN}{2\epsilon_{CTN}} r \\ F_{BOX}(r) &= -\frac{K_4}{\epsilon_{ox}} \frac{1}{r} \end{aligned} \quad (13)$$

where

$$K = \frac{V_G - V_{ch}}{\alpha} - \frac{qn_L n_1 (r_{n1}^2 - r_{o1}^2)}{2\alpha} \left[-\frac{r_{o1}^2}{r_{n1}^2 - r_{o1}^2} \frac{1}{\epsilon_{n1}} \ln \frac{r_{n1}}{r_{o1}} + \frac{1}{2\epsilon_{n1}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{o2}}{r_{n1}} + \frac{1}{\epsilon_{CTN}} \ln \frac{r_{CTN}}{r_{o2}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{BOX}}{r_{CTN}} \right] - \frac{qn_L CTN (r_{CTN}^2 - r_{o2}^2)}{2\alpha} \left[-\frac{r_{o2}^2}{r_{CTN}^2 - r_{o2}^2} \frac{1}{\epsilon_{CTN}} \ln \frac{r_{CTN}}{r_{o2}} + \frac{1}{2\epsilon_{CTN}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{BOX}}{r_{CTN}} \right] \quad (14)$$

$$K_1 = K - \frac{1}{2} qn_L n_1 (r_{n1}^2 - r_{o1}^2), \quad K_2 = K + \frac{1}{2} qn_L n_1 (r_{n1}^2 - r_{o1}^2) \quad (15)$$

$$K_3 = K + \frac{1}{2} qn_L n_1 (r_{n1}^2 - r_{o1}^2) - \frac{1}{2} qn_L CTN r_{o2}^2 \quad (16)$$

$$K_4 = K + \frac{1}{2} qn_L n_1 (r_{n1}^2 - r_{o1}^2) + \frac{1}{2} qn_L CTN (r_{CTN}^2 - r_{o2}^2) \quad (17)$$

$$\alpha = \frac{1}{\epsilon_{ox}} \ln \frac{r_{o1}}{r_{Si}} + \frac{1}{\epsilon_{n1}} \ln \frac{r_{n1}}{r_{o1}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{o2}}{r_{n1}} + \frac{1}{\epsilon_{CTN}} \ln \frac{r_{CTN}}{r_{o2}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{BOX}}{r_{CTN}} \quad (18)$$

Detailed Poisson Solution needed!

Detailed Poisson Solution

Used Poisson equation – electrical potential solutions

$$V_{o1}(r) = \frac{K_{o1}}{\epsilon_{o1}} \ln \frac{r}{r_{Si}}$$

$$V_{n1}(r) = \frac{K_{o1}}{\epsilon_{o1}} \ln \frac{r_{o1}}{r_{Si}} + \frac{K_{n1}}{\epsilon_{n1}} \ln \frac{r}{r_{o1}} + \frac{qn_{t_n1}}{4\epsilon_{n1}} (r^2 - r_{o1}^2)$$

$$V_{o2}(r) = V_{n1}(r_{n1}) + \frac{K_{o2}}{\epsilon_{ox}} \ln \frac{r}{r_{n1}}$$

$$i = 1$$

$$V_{CTN,1}(r) = V_{o2}(r_{o2}) + \frac{K_{CTN,1}}{\epsilon_{CTN}} \ln \frac{r}{r_{o2}} + \frac{qn_{t_CTN,1}}{4\epsilon_{CTN}} (r^2 - r_{o2}^2)$$

$$i = 2 \sim N$$

$$V_{CTN,i}(r) = V_{CTN,i-1}(r_{CTN,i-1}) + \frac{K_{CTN,i}}{\epsilon_{CTN}} \ln \frac{r}{r_{tox}} + \frac{qn_{t_CTN,i}}{4\epsilon_{CTN}} (r^2 - r_{CTN,i-1}^2)$$

CTN Partition in N

$$V_{BOX}(r) = V_{CTN,N}(r_{CTN,N}) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r}{r_{CTN,N}}$$

Now, K-constants can be determined from the boundary conditions

Detailed Poisson Solution

Boundary Conditions

$$\begin{aligned}
 \left. \begin{aligned}
 r &= r_{o1} & K_{o1} &= K_{n1} + \frac{1}{2} q n_{t_{n1}} r_{o1}^2 \\
 r &= r_{n1} & K_{n1} + \frac{1}{2} q n_{t_{n1}} r_{o1}^2 &= K_{o2} \\
 r &= r_{o2} & K_{o2} &= K_{CTN,1} + \frac{1}{2} q n_{t_{CTN,1}} r_{o2}^2 \\
 \\
 r &= r_{CTN,i-1} & K_{CTN,i-1} + \frac{1}{2} q n_{t_{CTN,i-1}} r_{CTN,i-1}^2 &= K_{CTN,i} + \frac{1}{2} q n_{t_{CTN,i}} r_{CTN,i-1}^2 \\
 \\
 r &= r_{CTN,N} & K_{CTN,N} + \frac{1}{2} q n_{t_{CTN,N}} r_{CTN,N}^2 &= K_{BOX}
 \end{aligned} \right\}
 \end{aligned}$$

Unknowns $K_{o1}, K_{n1}, K_{o2}, K_{CTN,1 \sim N}, K_{BOX}$: Total $N+4$

Above equations $r_{o1}, r_{n1}, r_{o2}, r_{CTN,1 \sim N}$: Total $N+3$

One more BC needed

Detailed Poisson Solution

Boundary Condition : Gate Voltage

$$V_{BOX}(r) = V_{CTN,N}(r_{CTN,N}) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r}{r_{CTN,N}}$$



$$r = r_{BOX}$$

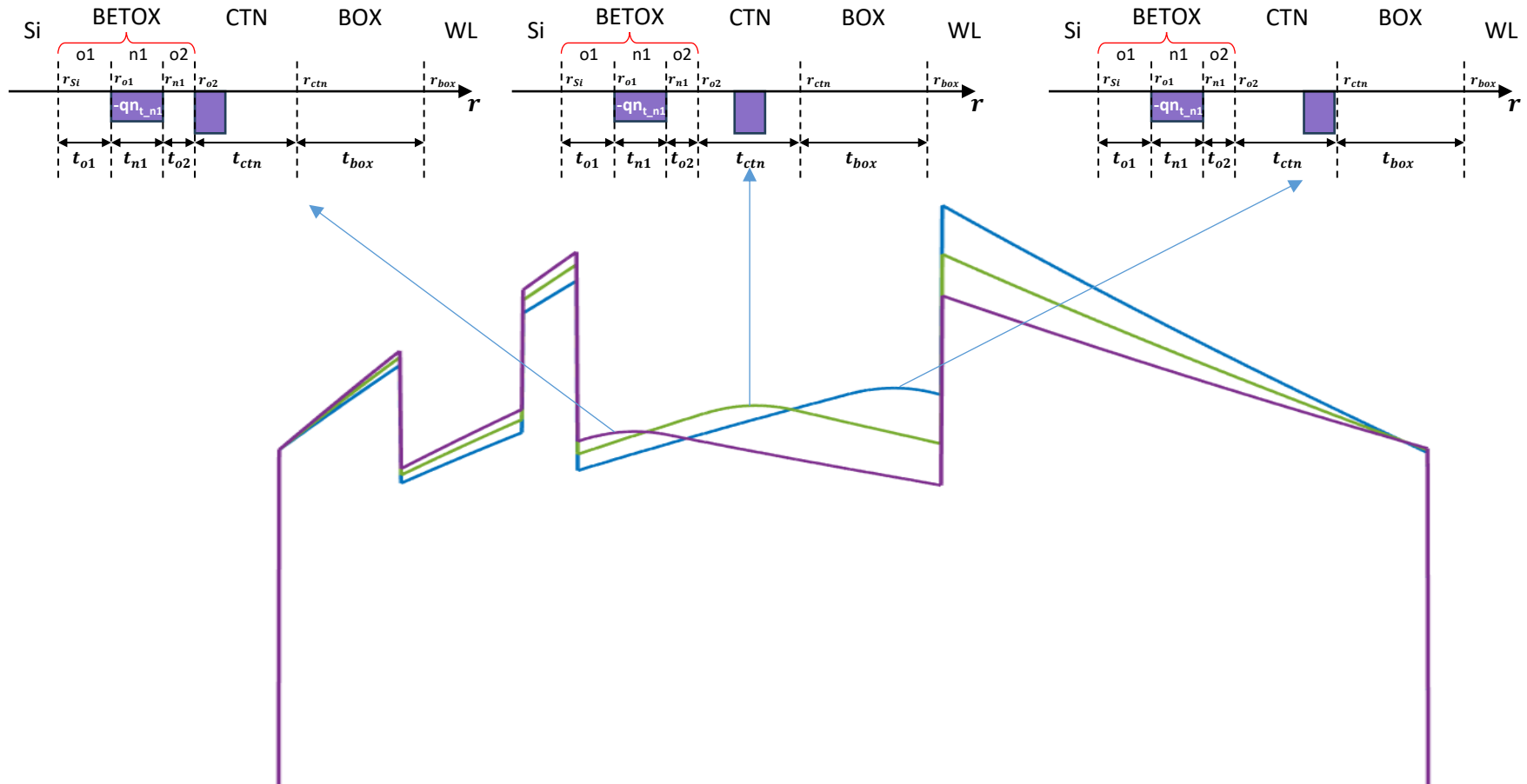
$$V_{BOX}(r_{BOX}) = V_{CTN,N}(r_{CTN,N}) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r_{BOX}}{r_{CTN,N}} = V_G$$

in result,

$$V_G = \frac{K_{o1}}{\epsilon_{o1}} \ln \frac{r_{o1}}{r_{Si}} + \frac{K_{n1}}{\epsilon_{n1}} \ln \frac{r_{n1}}{r_{o1}} + \frac{qn_{t,n1}}{4\epsilon_{n1}} (r_{n1}^2 - r_{o1}^2) + V_{n1}(r_{n1}) + \frac{K_{o2}}{\epsilon_{o2}} \ln \frac{r_{o2}}{r_{n1}} + \frac{K_{CTN,1}}{\epsilon_{CTN}} \ln \frac{r_{CTN,1}}{r_{o2}} \\ + \frac{qn_{t_CTN,1}}{4\epsilon_{CTN}} (r_{CTN,1}^2 - r_{o2}^2) + \sum_{i=2}^N \left\{ \frac{K_{CTN,i}}{\epsilon_{CTN}} \ln \frac{r_{CTN,i}}{r_{CTN,i-1}} + \frac{qn_{t_CTN,i}}{4\epsilon_{CTN}} (r_{CTN,i}^2 - r_{CTN,i-1}^2) \right\}$$

Now, unknowns can be solved

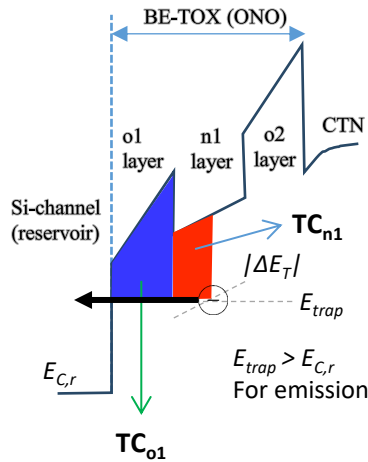
Poisson Solution Variations



Variations of trap distributions also can be discussed

CTN Detrapping Simulation Method

BETOX Detrapping Summary



➤ Emission rate fundamental equation: $e_{\text{TBT}} = \nu_{\text{TBT}} \times |\text{TC}|$

➤ Transmission coefficient with WBK approximation method

$$|\text{TC}| = \prod_{o1, n1} \exp\left(-\frac{4\sqrt{2m_{\text{ox},n}}}{3\hbar q} \cdot \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F_{\text{eq}}}\right)$$

➤ Attempt-to-escape frequency modeling

$$\nu_{\text{TBT}} = \int W(E) \cdot D(E) \cdot (1 - f_r(E)) dE$$

➤ $W(E)$: Transition rate between band and trap by constant perturbation

$$W(E) = \frac{2\pi}{\hbar} |\langle f | \hat{H}' | i \rangle|^2 \rho(E) = \frac{2\pi}{\hbar} |\Delta E_T|^2 V_T \delta(E - E_{\text{trap}}), \quad \hat{H}'(\mathbf{r}) = \begin{cases} -|\Delta E_T|, & \text{for } \mathbf{r} \in V_T \\ 0, & \text{otherwise} \end{cases}$$

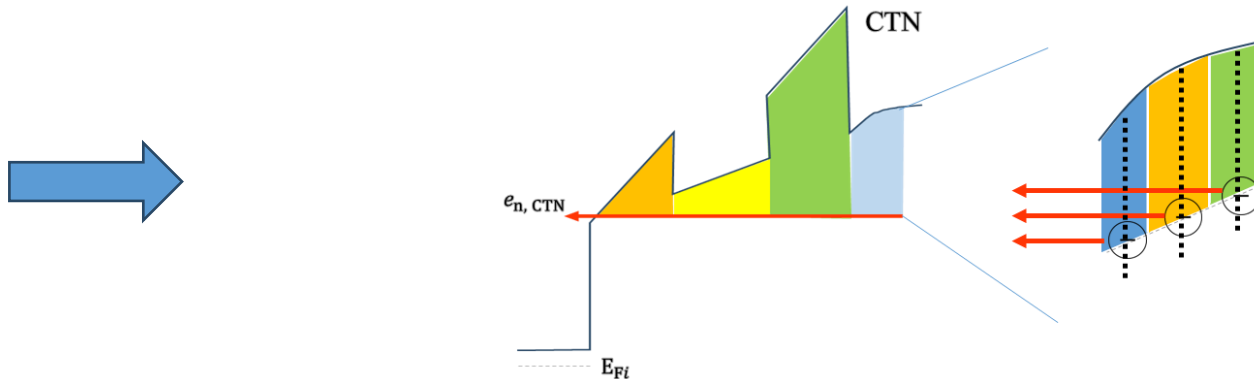
➤ $D(E)$: Density of state

$$D(E) = D_{\text{tun}} D_{\perp \text{tun}}(E) = 2 \cdot \frac{1}{4\pi^2} \frac{2m_0}{\hbar^2} \frac{\sqrt{2m_n^*}}{\hbar} \sqrt{E - E_{C,r}}$$

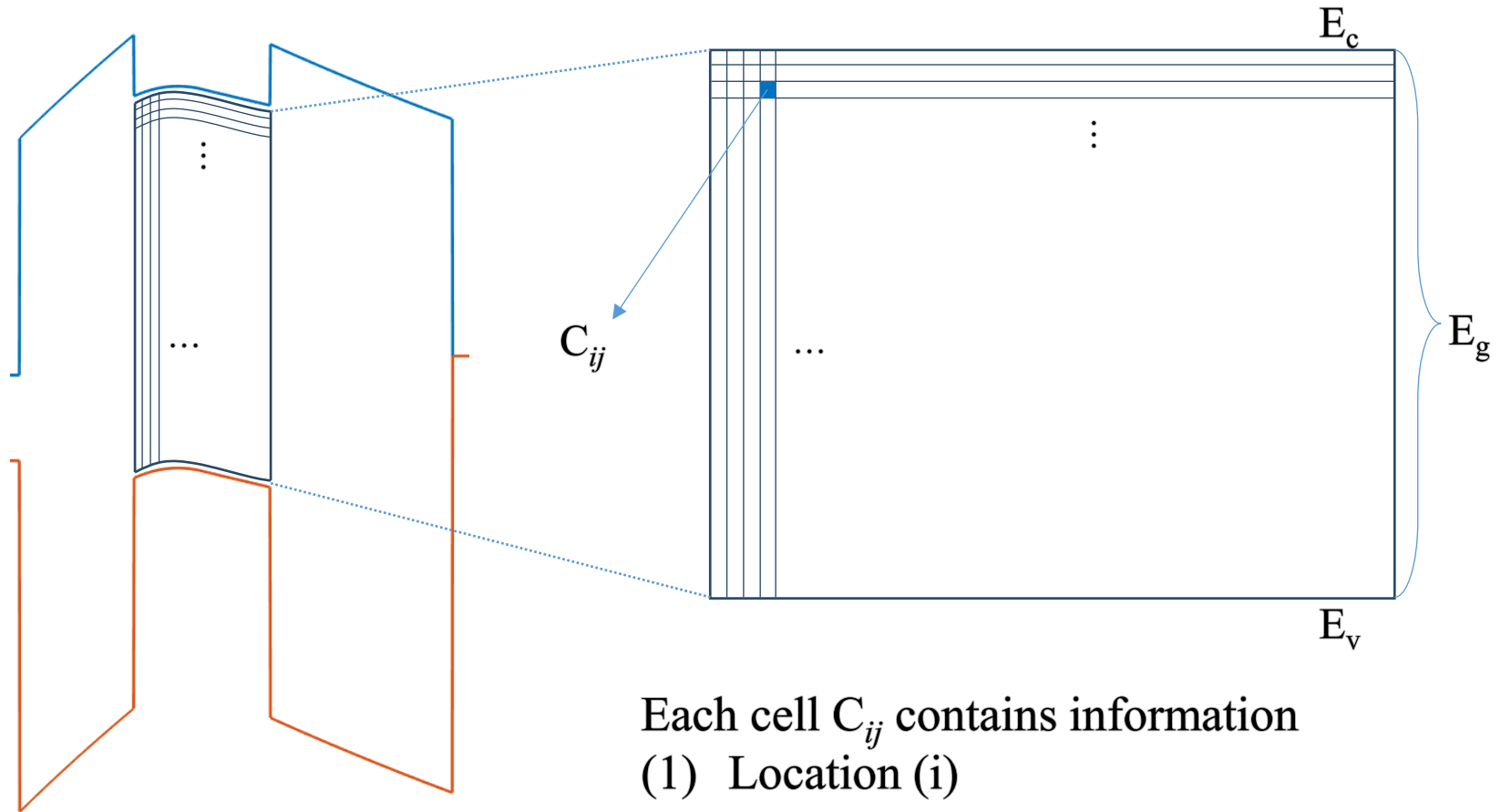
➤ Modeling result of tunneling emission rate

$$\nu_{\text{TBT}} = \frac{2\pi}{\hbar} |\Delta E_T|^2 V_T \times \frac{\sqrt{2m_n^*} m_0}{\pi^2 \hbar^3} \sqrt{E_{\text{trap}} - E_{C,r}}$$

CTN Detrapping Emission Modeling based on Detailed Poisson Solution



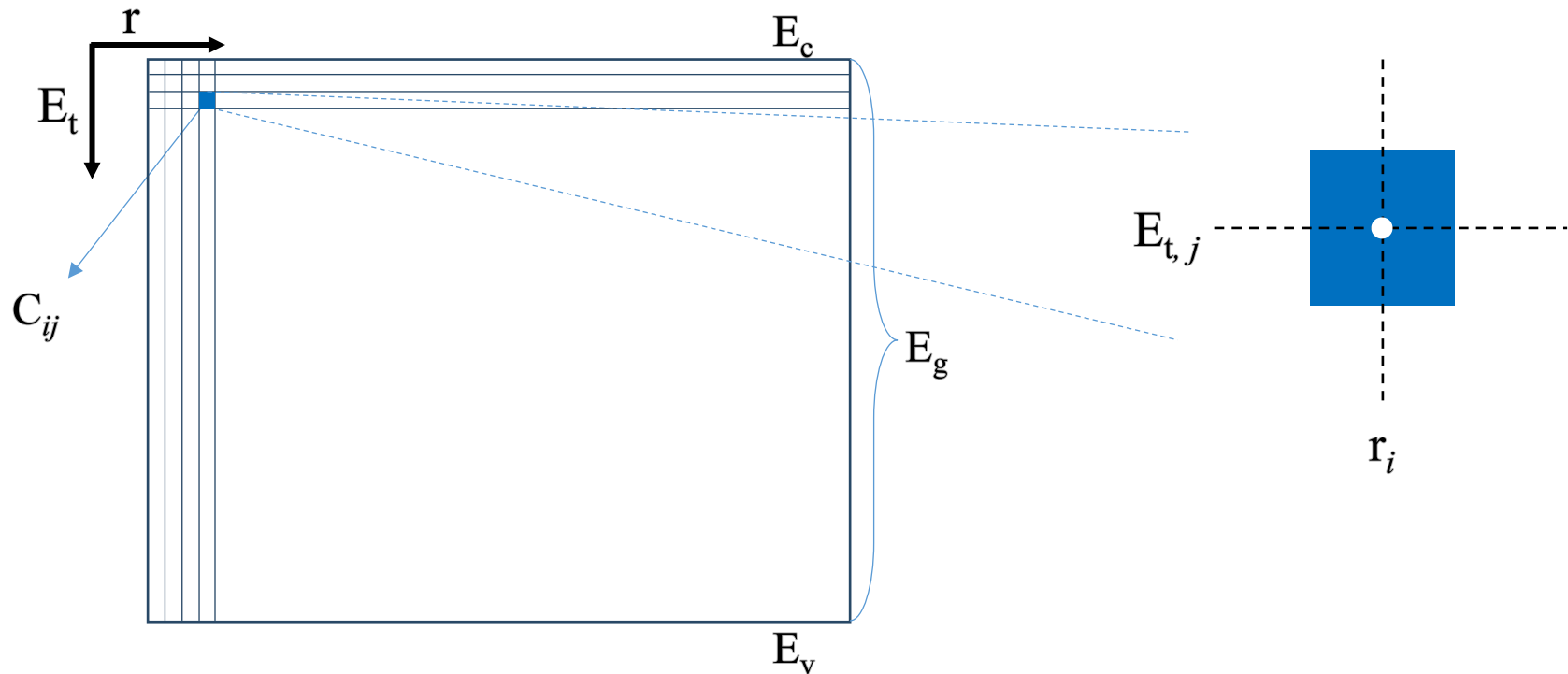
Cell Information Methodology



Each cell C_{ij} contains information

- (1) Location (i)
- (2) Trap depth (j)
- (3) Trap concentration
- (4) Emission rate to channel direction
- (5) Emission rate to wordline direction

Cell Information Methodology



- (1) Location (i)
- (2) Trap depth (j)

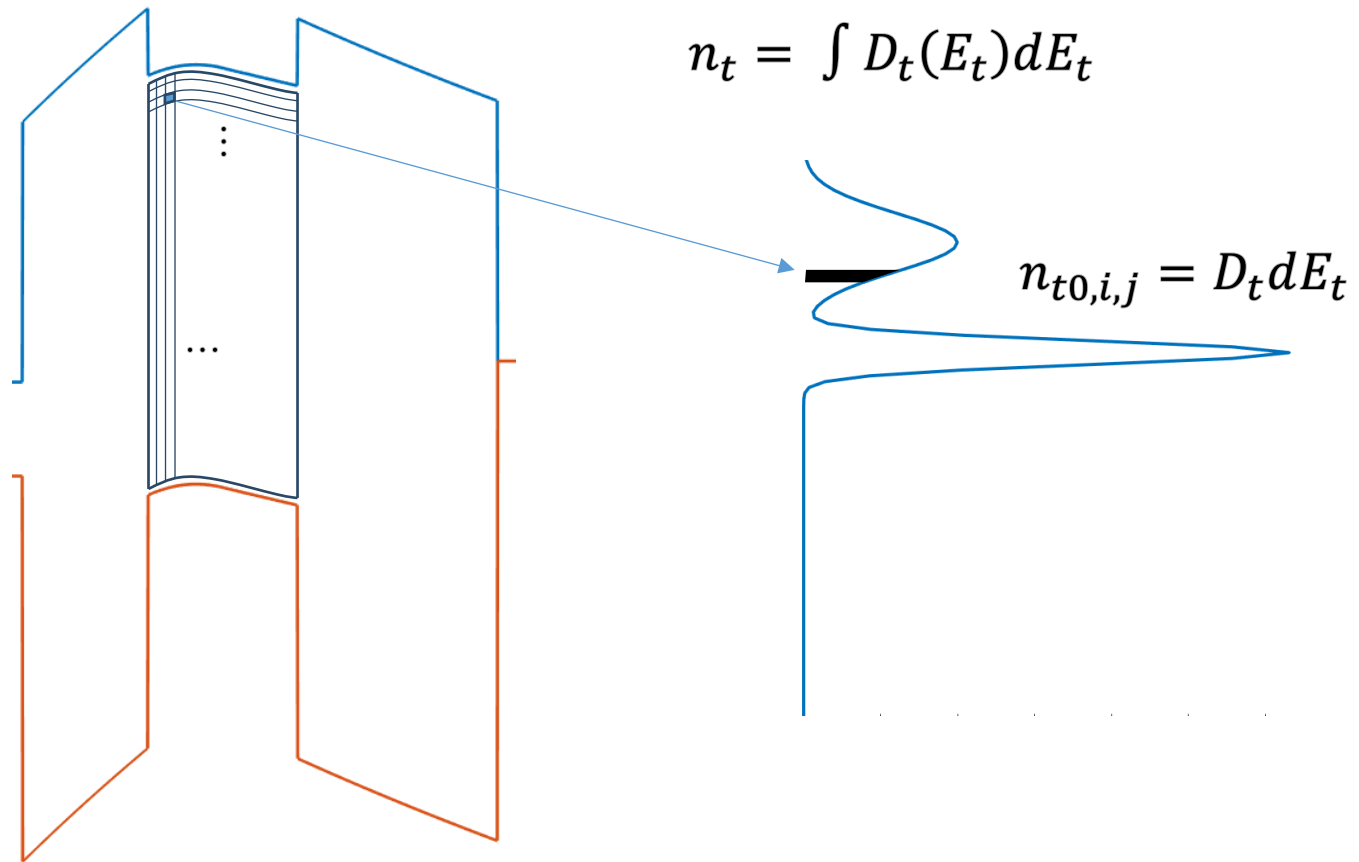
Used for cell monitoring

Location & trap depth saved at center point

Cell Information Methodology

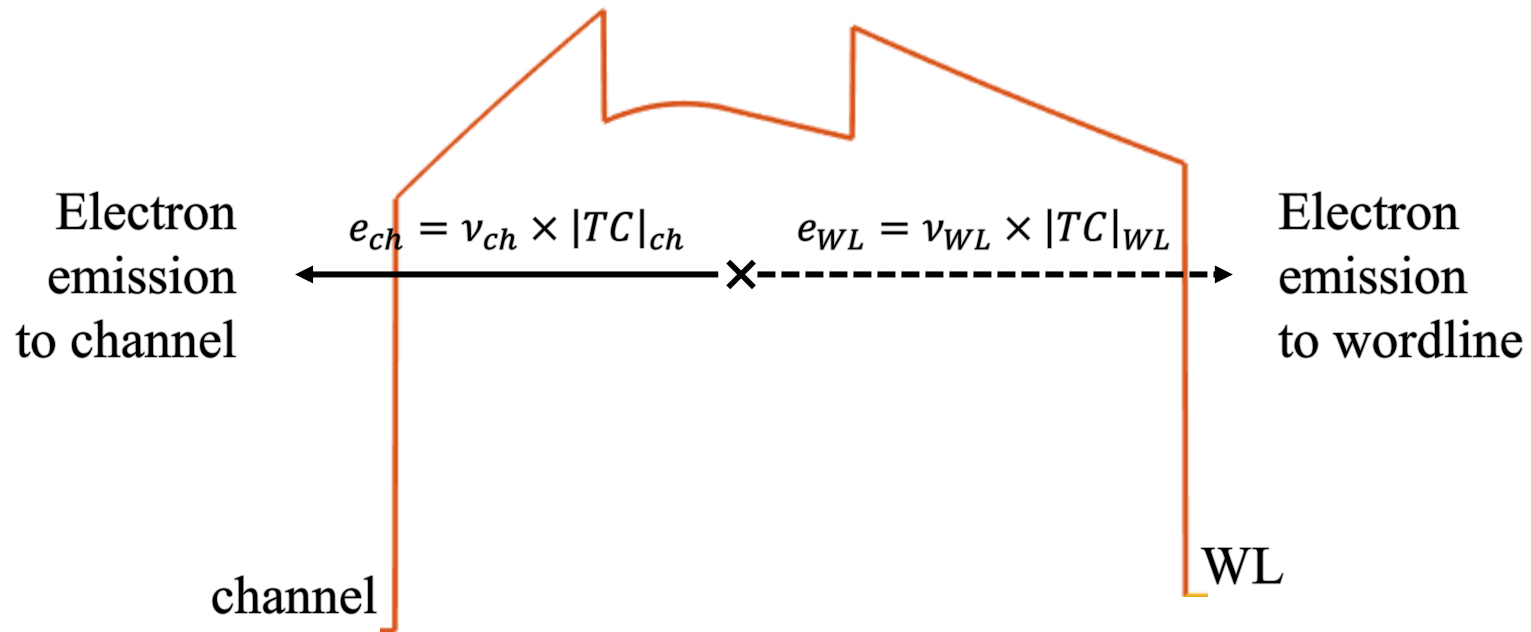
(3) Trap concentration

- local trap concentration based on location & trap depth



Cell Information Methodology

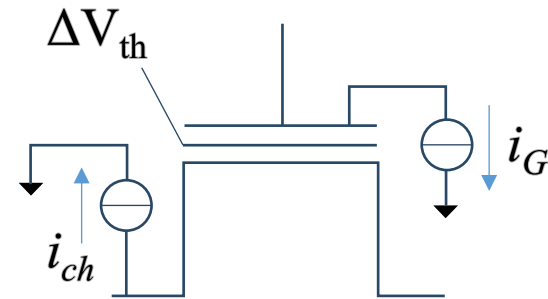
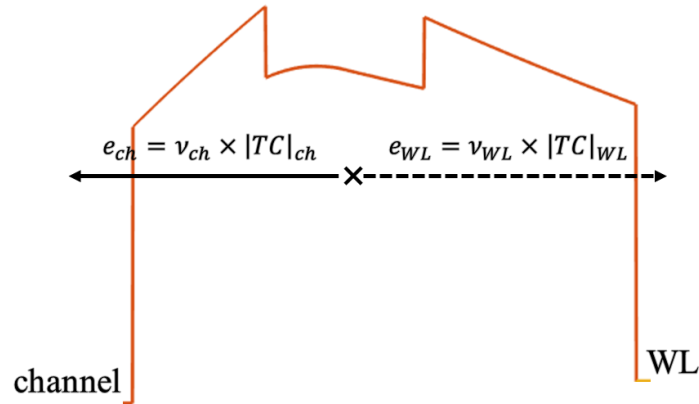
- (4) Emission rate to channel direction
- (5) Emission rate to wordline direction



$$n_t = \sum_{i,j} n_{t,i,j} = \sum_{i,j} n_{t0,i,j} \{ \exp(-e_{ch,i,j} t) + \exp(-e_{WL,i,j} t) \}$$

Equivalent Circuit

Equivalent circuit modeling as **dependent current source**



$$n_t = \sum_{i,j} n_{t,i,j} = \sum_{i,j} n_{t0,i,j} \{ \exp(-e_{ch,i,j} t) + \exp(-e_{WL,i,j} t) \}$$

$$i_{ch} = -q \frac{dn_{t,ch}}{dt} = -q \frac{d}{dt} \sum_{i,j} n_{t0,i,j} \exp(-e_{ch,i,j} t) = q \sum_{i,j} e_{ch,i,j} n_{t0,i,j} \exp(-e_{ch,i,j} t)$$

$$i_G = -q \frac{dn_{t,G}}{dt} = -q \frac{d}{dt} \sum_{i,j} n_{t0,i,j} \exp(-e_{G,i,j} t) = q \sum_{i,j} e_{G,i,j} n_{t0,i,j} \exp(-e_{G,i,j} t)$$

=> Modeling by leakage current is on progress

Summary

Detailed Poisson equation solution solved for CTN trap distribution

Used Poisson equation

$$V_{o1}(r) = \frac{K_{o1}}{\epsilon_{o1}} \ln \frac{r}{r_{Si}}$$

$$V_{n1}(r) = \frac{K_{o1}}{\epsilon_{o1}} \ln \frac{r_{o1}}{r_{Si}} + \frac{K_{n1}}{\epsilon_{n1}} \ln \frac{r}{r_{o1}} + \frac{qn_t n_1}{4\epsilon_{n1}} (r^2 - r_{o1}^2)$$

$$V_{o2}(r) = V_{n1}(r_{n1}) + \frac{K_{o2}}{\epsilon_{ox}} \ln \frac{r}{r_{n1}}$$

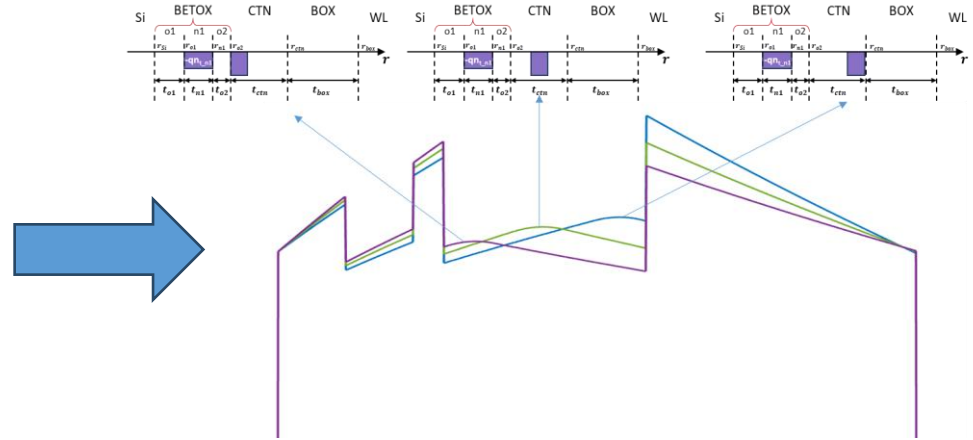
$$i = 1$$

$$V_{CTN,1}(r) = V_{o2}(r_{o2}) + \frac{K_{CTN,1}}{\epsilon_{CTN}} \ln \frac{r}{r_{o2}} + \frac{qn_{t,CTN,1}}{4\epsilon_{CTN}} (r^2 - r_{o2}^2)$$

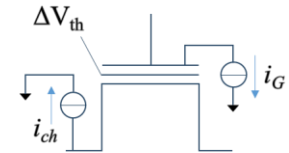
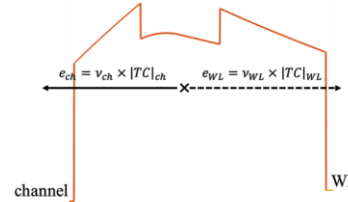
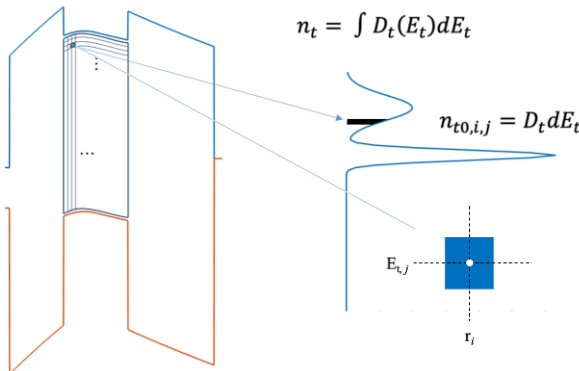
$$i = 2 \sim N$$

$$V_{CTN,i}(r) = V_{CTN,i-1}(r_{CTN,i-1}) + \frac{K_{CTN,i}}{\epsilon_{CTN}} \ln \frac{r}{r_{tox}} + \frac{qn_{t,CTN,i}}{4\epsilon_{CTN}} (r^2 - r_{CTN,i-1}^2)$$

$$V_{BOX}(r) = V_{CTN,N}(r_{CTN,N}) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r}{r_{CTN,N}}$$



Cell information methodology / Modeling leakage current



$$n_t = \sum_{i,j} n_{t,i,j} = \sum_{i,j} n_{t0,i,j} \{ \exp(-e_{ch,i,j} t) + \exp(-e_{WL,i,j} t) \}$$

$$i_{ch} = -q \frac{dn_{t,ch}}{dt} = -q \frac{d}{dt} \sum_{i,j} n_{t0,i,j} \exp(-e_{ch,i,j} t) = q \sum_{i,j} e_{ch,i,j} n_{t0,i,j} \exp(-e_{ch,i,j} t)$$

$$i_G = -q \frac{dn_{t,G}}{dt} = -q \frac{d}{dt} \sum_{i,j} n_{t0,i,j} \exp(-e_{G,i,j} t) = q \sum_{i,j} e_{G,i,j} n_{t0,i,j} \exp(-e_{G,i,j} t)$$